Problem Management

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1. Description

1.1. Overview

Problem Management is the process responsible for eliminating errors from the environment and reducing the incident volumes experienced by the users. Problem Management is also responsible for developing and publishing information that can assist in the rapid resolution of incidents. This information includes workarounds, publication of Known Errors, diagnostic scripts and other information that can assist with the rapid resolution of the incidents.

1.2. Purpose

The objective of problem management is to minimize the adverse impact to business, due to incidents and problems caused by errors within the IT Infrastructure and applications. Problem Management does this by fixing root causes, thus reducing future incidents.

To achieve this goal, Problem Management performs Root Cause Analysis (RCA) of problems to determine where the error is. Problem Management then converts this problem to a Known Error and manages the Known Error through resolution.

1.3. Business Value

Business values the Problem Management process through the reduction in Known Errors. IT services have higher availability, users and IT staff increase productivity, and there is a reduction in the costs of resolving repeat incidents.

Problem Management is also responsible for documenting workarounds to improve the resolution times of incidents. Proper Problem Management reduces the costs of these workarounds or fixes that do not work.

1.4. Key Concepts

Problem – A problem is defined as the unknown cause of one or more incidents. Problems, once diagnosed and the root cause is found, result in the documentation of Known Error.

Known Error - A Known Error is defined as a fault in the infrastructure for which a permanent solution or temporary workaround has been found and documented.

Workaround – A workaround is a pre-defined technique for reducing or eliminating the impact of an incident or a problem for which a full resolution in not available yet.

Known Error Database – When the root cause of problems is determined, a Known Error is created and associated with its workaround or permanent fix. This information is stored in a Know Error Database.

The goal of Problem Management process is to prevent incidents from occurring, as well as minimize the impact of incidents that cannot be prevented. The Problem Management process is depicted in the attachment (refer section 3.1.) and consists of the following procedures:

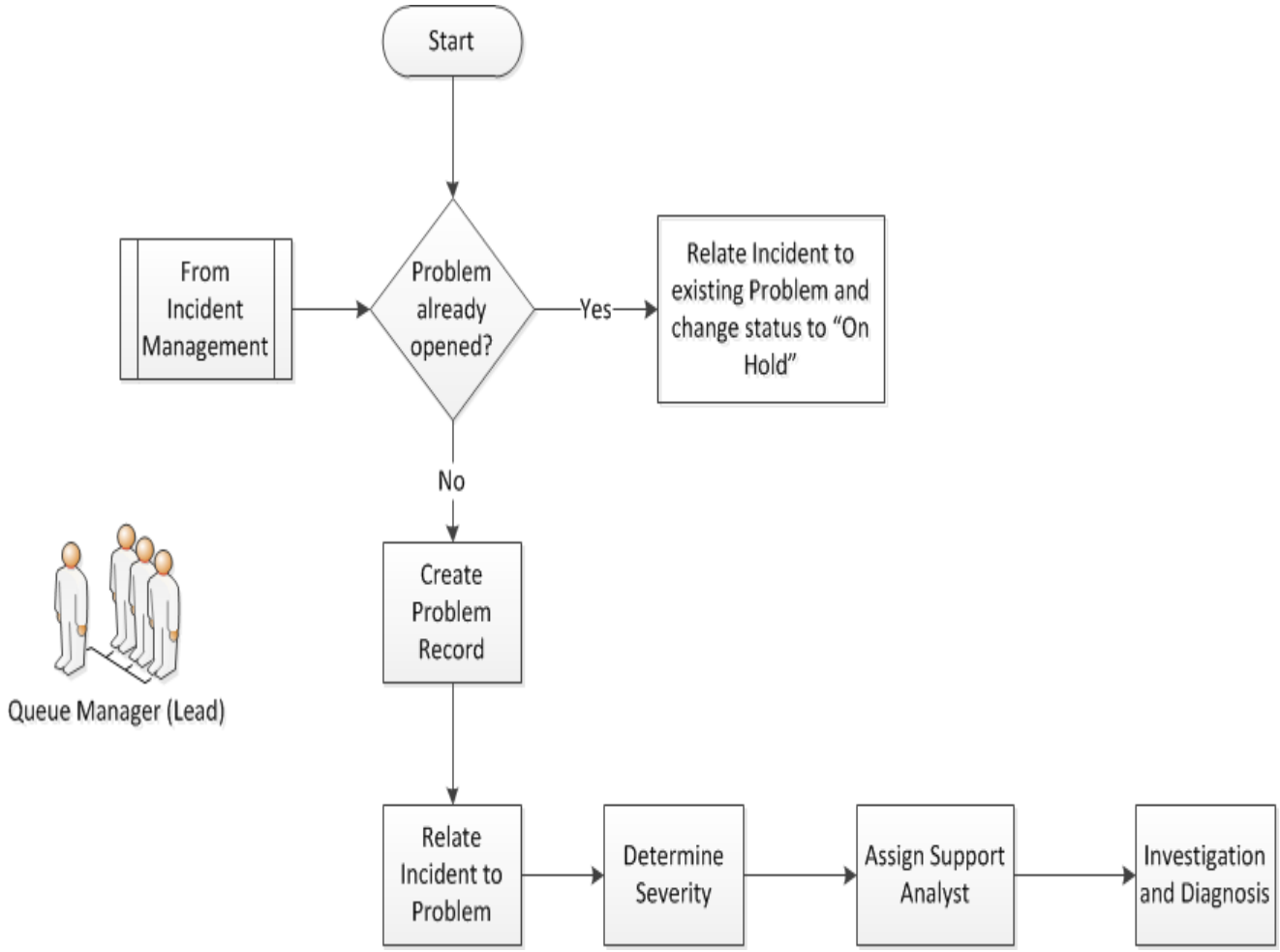
1. Incident Request Review
2. Incident Request Escalation
3. Problem Determination and Review
4. Problem Closure Incident Request Escalation

2. Problem Management Procedures

2.1. Incident Request Review

The Level 1 Support escalates incidents that are caused by a common problem to the Queue Manager (Level 2 Support Lead). The Queue Manager (Level 2 Support Lead) reviews the incidents and determines if Problem Management processes need to be invoked. The Queue Manager (Level 2 Support Lead) creates a Problem and links the related incidents. The Queue Manager (Level 2 Support Lead) can also invoke Problem Management processes for Severity 1 incidents. The Queue Manager (Level 2 Support Lead) then assigns the Problem to the Level 2 Support Analyst for investigation of the Root Cause.

The Review Incident Request procedure is shown below.

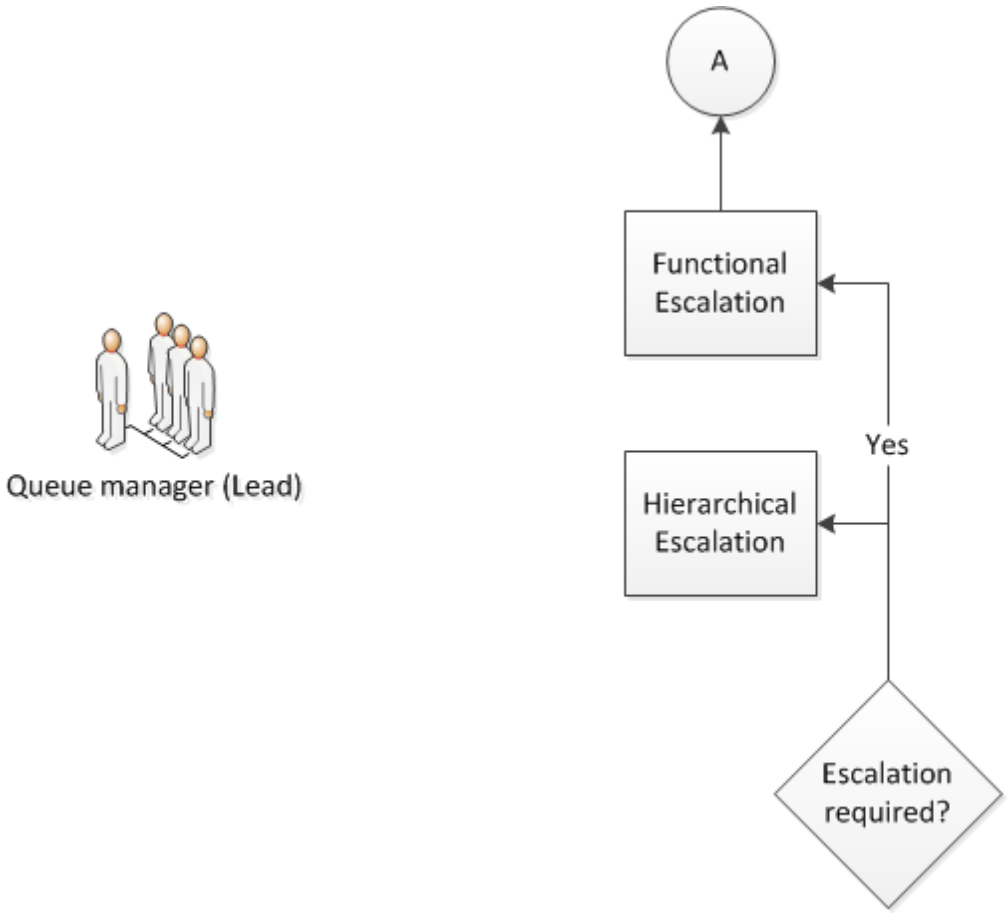


2.2. Problem Request Escalation

The Queue Manager (Level 2 Support Lead) evaluates the best solution to resolve the problem. He escalates the problem to the Project Manager if it needs to be recovered by the Level 3 Support. He performs risk and impact assessment if Change Management is required.

2.2.1. Functional and Hierarchical Escalations

The Incident Request Escalation procedure is shown below.



Functional and Hierarchical Escalations

Functional Escalation: *At times Level 2 Support or anyone who is investigating a problem may not have the ability to determine the root cause and provide a permanent fix. This may require the problem to be escalated to Level 3 with more specialized knowledge of that service or component. This is known as functional escalation.*

Hierarchic Escalation: *A problem may be escalated to a supervisor, manager, director, or higher to notify them of the problem and possible impact, as well as to obtain resources to resolve the problem. This is known as hierarchic escalation.*

The table below depicts the Description, Entry/Exit criteria, Environment & Tools, Templates, Actors and Responsibilities for each activity of the procedure.

Section	Description	Entry Criteria	Exit Criteria	Environment & Tools	Templates	Actors
Functional Escalation	Level 2 Support is not able to determine a root cause to a problem, then it is escalated to Level 3 Support with more specialized skill	Root cause couldn't be determined	Level 3 informed and issue being worked upon	NA	NA	Level 2 Support Project Manager, Level 3 Support
Hierarchic Escalation	Level 2 Support not able to determine a root cause for a problem and more specialized skill is required, then hierarchic escalation gets triggered	Root cause couldn't be determined and more specialized resources needed to work on the problem	Product Owner informed (*refer Product Owner Contact List from Appendix*)	NA	NA	Level 2 Support Project Manager, Client Director and any other higher official

2.2.2. Problem Determination (Level 3 Support)

When Level 2 Support is not able to determine root cause to a problem, it is escalated to the Level 3 Support. The Problem Determination by Level 3 Support is shown below.

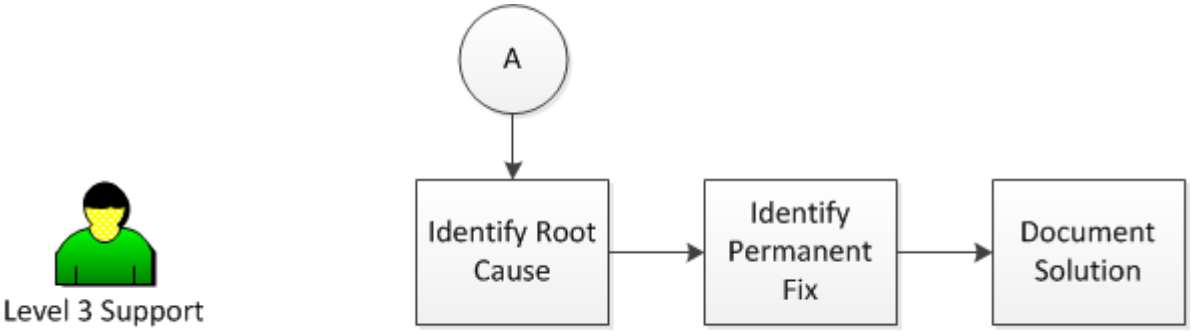


Figure 1: Problem Determination (Level 3 Support)

2.3. Problem Determination and Review

For Problems that have related incidents, the Level 2 Support Analyst identifies a temporary work around. He updates the problem record with the details of the work around. The Level 2 Support Analyst performs Root Cause Analysis and evaluates options to identify a permanent fix. If the permanent fix does not require Change Management then the Level 2 Support Analyst implements

the permanent fix. If implementation of the permanent fix involves degradation of application performance during business hours or involves changes to the application functionality then the Level 2 Support Analyst escalates the Problem to the Queue Manager (Level 2 Support Lead) for invoking the Change Management processes. If the Level 2 Support Analyst is unable to determine a permanent fix then he/she submits the investigation findings to the Queue Manager (Level 2 Support Lead).

The Queue Manager (Level 2 Support Lead) reviews the history log for problem analysis. If the analysis is not satisfactorily done then Queue Manager (Level 2 Support Lead) re-assigns the problem for further investigation. The Queue Manager (Level 2 Support Lead) evaluates if Change Management is required for the implementation of the permanent fix. If Change Management processes are required then the Queue Manager (Level 2 Support Lead) creates a Known Error record before invoking the Change Management processes. If the Queue Manager (Level 2 Support Lead) feels that Change Management processes are not required then he re-assigns the problem for further analysis.

The Problem Determination and Review procedure is shown below.

Section	Description	Entry Criteria	Exit Criteria	Environment & Tools	Templates	Actors	Responsibilities
Identify Root Cause	Level 2 Support Analyst investigates and identifies a root cause to a problem	Investigation and Diagnosis done and no escalations needed	Root Cause identified	Dev/Test	Cut-Over checklist	Level 2 Support Analyst	• Level 2 Support Analyst investigates and identifies the root cause to a problem
Identify Permanent Fix	Level 2 Support Analyst identifies a permanent fix once the root cause to a problem is identified	Root Cause identified	Permanent fix identified	Dev/Test	NA	Level 2 Support Analyst	• Level 2 Support Analyst works on a problem to identify permanent fix once the root cause it identified
Implement Permanent Fix	Level 2 Support Analyst implements the permanent fix identified in case there is no code change required	Permanent fix identified and no code change required	Permanent fix implemented	Production	NA	Level 2 Support Analyst	• Level 2 Support Analyst implements the fix once he identifies the permanent fix
Create Problem Resolution Record	Once the permanent fix is implemented, a problem resolution record is created	Permanent fix implemented	Problem resolution record created	Ticket entering tool	Problem resolution record	Level 2 Support Analyst	• Level 2 Support Analyst creates a problem resolution record once he implements the permanent fix
Identify and deploy Workaround	Level 2 Support Analyst determines the appropriate workaround for the problems. These workarounds are documented and recorded in the Known Error database	Code change required, No Emergency Change needed	Workaround identified and deployed. Level 1 Support notified with the workaround for future similar occurrence	Production	Known Error record	Level 2 Support Analyst	• Level 2 Support Analyst identifies and deploys the appropriate workaround and notifies the Level 1 Support through Incident Management process

Update Product Backlog: Prioritization	An enhancement is logged in JIRA and status marked as “Backlog”	Code change required	Backlog prioritized and ready for construction and testing	JIRA/ Equivalent	Design Document, Unit Test plan	Queue Manager (Level 2 Support Lead)	• Queue Manager (Level 2 Support Lead) creates a request in JIRA and mark as backlog • Work prioritized
Emergency Change Management	Queue Manager (Level 2 Support Lead) invokes Emergency Change Management in order to prevent unacceptable loss or impact to the business	Emergency as there is specific business impact requirements to be considered	Change implemented and a change implementation note is submitted with the Change Advisory Board (CAB)	NA	RFC, Change Implementation Note	Queue Manager (Level 2 Support Lead), Change Advisory Board (CAB) members	• Queue Manager (Level 2 Support Lead) invokes Emergency Change Management depending on specific business requirements to be considered • Change Advisory Board (CAB) evaluates and approves request for change
Code Construction and Testing	Once prioritized, Level 2 Support Analyst starts development and testing	Prioritization done and estimates provided	Construction and testing done successfully	Dev/Test	Design Document, Unit Test Cases	Level 2 Support Analyst	• Level 2 Support Analyst develops code and tests the modification
Create RCA Report	Level 2 Support Analyst creates an RCA report, which has a description of the reported incident, a causation statement indicating the reasons why the incident occurred, based on the findings from RCA investigation, and recommendations by the RCA team of changes or improvements relating to a procedure or practice arising out of the incident.	Permanent fix is identified	RCA report ready for review	NA	RCA Template	RCA team, Level 2 Support	• Level 2 Support Analyst creates an RCA report and submits it for review

RCA Review	RCA team reviews the RCA report and provides approval	RCA report submitted for review	RCA report reviewed and approved	NA	NA	RCA template	• RCA team reviews the RCA report and approves it

2.4. Problem Closure

The Queue Manager verifies if the permanent fix solved the original problem. If the original problem is resolved, then he closes the problem and the Known Error record. If however, the original problem is not fixed then he evaluates possible alternative solutions. He reassigns the problem if alternative solutions are available. The Queue Manager periodically monitors the problem for new options if alternative solution is not available.

Section	Description	Entry Criteria	Exit Criteria	Templates	Actors	Responsibilities
Problem Closure	The Queue Manager verifies if the permanent fix solved the original problem and closes the problem and the Known Error record	Problem resolution record created	Resolution verified and all possible documents are completed	NA	Queue Manager (Level 2 Support Lead)	• Queue Manager (Level 2 Support Lead) verifies if the solution provided fixed the original problem and all documents are completed related to the problem

3. Tools and Templates

Project teams shall use the tools and templates listed in this section.

Tools
USTOptim - PMTool (ustpmtweb.azurewebsites.net)
Knowledge Base

Templates
Runbook Template

5. References

Guidelines & User Manual
Incident Management

6. Revision History

Version	Release Date	Description of Changes	Author/Reviewer/Approver (Name & Role)
1.0	19-Feb-19	Initial Release	Rejith Raveendran Nair, EPEX / Pradeep Kumar Pullathil, Head EPEX / Pradeep Kumar Pullathil, Head EPEX /
R2-V1.0	03-Mar-2022	Template updated with the new UST Brand	Pragya Pandey / Srinivasan Nagaraj
R2-V1.1	Mar-25-22	Branding change implemented	Pragya Pandey/Srinivasan Nagaraj
R2-V1.2	09-Feb-26	Sec 2.1 updated	Meenakshi J / Rajesh Sreekumaran

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